

Module 03: Perception — Robot Senses

Learning Objectives

1. Understand that **Sensors** are how robots feel the world.
2. Compare Human Senses (Eyes, Ears) to Robot Sensors (Cameras, Mics).
3. Learn that robots need good data to make good decisions.

Introduction: The Robot Senses

How do you know if a wall is in front of you? You see it with your eyes. How does a robot know? It uses a **Sensor**.

- **Camera:** Sees light and color.
- **Microphone:** Hears sounds.
- **Bumper:** Feels a crash.
- **Thermometer:** Feels hot and cold.

Without sensors, a robot is blind and deaf!

Key Concepts

1. The Camera (Robot Eye)

A camera takes pictures really fast. The robot looks at the pixels. "I see a red blob. That is a ball." "I see a green blob. That is a tree."

2. The Microphone (Robot Ear)

Robots can listen to commands. When you say "Hey Siri" or "Alexa," a machine is listening! It turns sound waves into code.

3. Invisible Senses

Robots have superpowers! They can sense things humans can't.

- **Lidar:** Seeing with lasers!
- **Infrared:** Seeing heat (Night Vision).
- **Magnetism:** Knowing which way is North.

Activities

Activity 1: The Blind Robot

One person is the Robot (Blindfolded). One person is the Sensor (The Guide). The Guide must whisper what they see to help the Robot walk safely. "Wall ahead. Stop." This shows how much the Robot trusts the Sensor!

Activity 2: Sensor Scavenger Hunt

Look around your house or school. Find 3 Machines with sensors.

1. **Smoke Alarm:** Smells smoke!
2. **Automatic Door:** Sees you coming!
3. **Thermostat:** Feels the heat!

Summary

Perception is how robots understand the world. They use sensors to gather data, just like we use our five senses. Some robots can even sense things we can't!

References

- *Robots, Robots Everywhere!* by Sue Fliess
- *Sensors: Eyes and Ears of a Robot* (Kids Engineering Video)